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Amorphous Engineering of Scalable Metal-Organic Framework-Derived Electrocatalyst for Highly Efficient Oxygen Evolution Reaction

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The engineering of amorphous metal-organic frameworks (MOFs) offers potential opportunities for the construction of electrocatalysts for efficient oxygen evolution reaction (OER). Herein, highly efficient OER performance and durability in alkaline electrolyte are discovered for MOF-derived amorphous and porous electrocatalysts, which are synthesized in a brief procedure and can be facilely produced in scalable quantities. The structural inheritance of MOF amorphous catalysts is significant for the retention of catalytic sites and the diffusion of electrolytes, and the presence of Fe sites can change the electronic structure and effectively control the adsorption behavior of important intermediates, accelerating reaction kinetics. The obtained amorphous A-FeNi can be transformed from FeNi-MOF effortlessly and instantly, and it only needs low overpotentials of 152 and 232 mV at 10 and 100 mA cm^{-2} with a Tafel slope of 17 mV dec^{-1} in 1 M KOH for OER. Moreover, A-FeNi possesses high corrosion resistance and durability, therefore A-FeNi can work continually for at least 400 h at 100 mA cm^{-2}. This work may pave a new avenue for the design of MOFs-related amorphous electrocatalyst.

1. Introduction

As global energy demand increases, energy reserves are dwindling and energy extraction is becoming more expensive and difficult.[1–3] One of the key measures to address energy depletion and the resulting environmental devastation is to exploit sustainable new energy sources including hydrogen energy, solar energy, wind energy, etc.[4–6] Electrocatalytic oxygen evolution reaction (OER) is important for environmental protection, energy conversion, and the preparation of high purity hydrogen and oxygen.[7–10] However, high energy barriers need to be overcome during OER process due to its complicated steps. Therefore it is critical to develop highly efficient and persistent catalysts that can reduce energy losses.[11] Moreover, new concepts of facile and scalable fabrication for electrocatalysis are also highly required.

Metal-organic frameworks (MOFs), as a new type of organic-inorganic hybrid materials, have captured extensive interest due to their explicit composition and adjustable coordination.[12–14] MOFs in crystalline state have been extensively studied in various field, ranging from gas adsorption and energy conversion to catalysis,[15–20] of which electrocatalysis is a hot research topic. Amorphous MOFs (AMOFs) or MOFs-derived amorphous materials have recently been proven to be a new class of promising catalysts.[21–23] Compared with conventional crystalline MOFs, amorphous MOFs-based catalysts tend to have more catalytic active sites and better corrosion resistance.[24,25] Additionally, due to the easier composition and milder synthesis condition, amorphous materials are expected to achieve higher electrochemical activity.[26–30] However, amorphous materials are challenging to be characterized for the following two main reasons: conventional X-ray sources with low energy levels make it difficult to accurately resolve the precise structure of amorphous materials; and traditional wave spectral characterization is deficient in local structure detection.[31,32] Therefore, there are not yet sufficient studies on amorphous MOFs-based catalysts. Furthermore, the synthesis process of many MOFs or MOFs-derived materials requires relatively high temperature, high pressure, and long time. The development of MOFs with facile synthesis process and scalable quantity can reduce both energy consumption and time cost, and contribute to the realization of industrial fabrication.

Herein, an in situ, facile, and scalable synthesis strategy for MOF-derived amorphous catalyst denoted as A-FeNi was proposed for highly efficient OER. FeNi-MOF was prepared using

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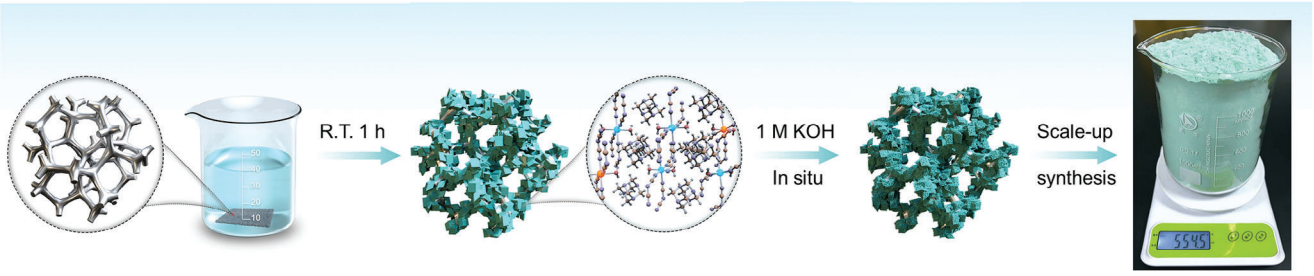


Figure 1. The schematic illustration for the preparation of FeNi-MOF and A-FeNi with a photograph of A-FeNi for scale-up synthesis.

a brief method and mild conditions to obtain a robust loading on Ni foam substrate. After preparation of the FeNi-MOF loading on Ni foam at room temperature as a composite electrode, the electrocatalytically active A-FeNi loading on Ni foam was in situ formed instantaneously upon immersion of the precursor FeNi-MOF electrode into 1 M KOH electrolyte just before OER process. The obtained A-FeNi exhibited a similar morphology to the original FeNi-MOF and a higher coordination number proven by X-ray absorption fine structure (XAFS) results. The excellent performance of A-FeNi may be derived from the ligand and coordination structure inherited from FeNi-MOF. Density function theory (DFT) calculation revealed that Ni site in A-FeNi performed a critical role in the first step of OER process, while Fe site acted as primary active site in the next three steps. The obtained A-FeNi exhibited top OER performance and outstanding durability, demonstrating the importance of MOFs-related amorphous catalysts for potential industrial applications.

2. Results and Discussion

FeNi-MOF was uniformly and securely loaded on the nickel foam in a relatively short time of 1 h at room temperature (R.T., 25 °C) with a loading amount of ≈5 mg cm^{-2} as schemed in **Figure 1**. The obtained FeNi-MOF showed a blue-green color as shown in Figure S1 (Supporting Information). It has been shown that other Ni-based MOFs (including CoNi-MOF and MnNi-MOF etc.) can also be prepared by this method and they were synthesized for comparison. Besides, these Ni-based MOFs can be firmly loaded onto other substrates including carbon paper and carbon cloth, indicating the good universality of the fabrication method. In the process of amorphization, the hydrolysis process became facilitated due to the strong basic of the electrolyte. OH^- ions of KOH electrolyte acted as a strong nucleophile and attracted partially positive metal ions. Metal-organic bonds were cleaved and ligands were partially removed. Hydrolysis caused a disruption of the crystalline structure of the original MOFs, leading to the breakdown of partial ligands and the liberation of metal sites. Moreover, inductively coupled plasma mass spectrometry (ICP-MS) was conducted for the electrolyte to demonstrate that metal ions are able to remain in the original MOFs during hydrolysis process. Five milligrams of FeNi-MOF powder was immersed in 100 mL 1 M KOH and then ICP-MS was conducted for the electrolyte after membrane filtration. The elemental contents for Fe and Ni were tested to be 13.8 and 2.5 ppb. A very small amount of metal leached out into the electrolyte during the amorphous transformation of A-FeNi. Thus, metal leakage is negligible during the hydrolysis process. The proportion of two metals in the catalyst was obtained by ICP-MS as listed in Table S1 (Supporting Information), and the proportion of Fe to Ni for A-FeNi was characterized to be 0.34:0.66. Moreover, FeNi-MOF and A-FeNi can be facilely prepared on a large scale in powder form (500 g for A-FeNi) as shown in Figure 1. When the MOFs loaded on Ni foam were emerged into 1 M KOH electrolyte as working electrode, the MOFs in situ transformed to amorphous catalysts instantaneously. Since the conversion process is fast and in situ, OER can be carried out immediately. Moreover, in order to investigate the difference between amorphous A-FeNi and its crystalline state catalyst, A-FeNi was annealed to obtain crystalline catalyst denoted as C-FeNi.[30]

Microscopically, FeNi-MOF exhibited a regular orthogonal hexahedron morphology with smooth surface as shown in the scanning electron microscopy (SEM) and transmission electron microscopy (TEM) images in **Figure 2**A,C. When the micron FeNi-MOF was in situ converted into an amorphous catalyst in the 1 M KOH electrolyte, the smooth surface of the original FeNi-MOF became rough and porous as shown in Figure 2B,D. It can also be observed that the hexagonal morphology of the precursor FeNi-MOF was roughly maintained in A-FeNi. As illustrated in high-magnification TEM image inset in Figure 2D, the surface of A-FeNi had a multi-level structure consisting of stacked ultra-small nanosheets. This structure may facilitate the diffusion length of electrolyte and the transport of electrons. The SEM images of Ni-MOF and A-Ni are displayed in Figures S2 and S3 (Supporting Information), illustrating that the morphology and change pattern are basically the same as bimetallic A-FeNi. The amorphous diffraction ring in Figure 2E demonstrated that there is no crystal phase in the structure of A-FeNi. As shown in Figures S4 and S5 (Supporting Information), the porous surface disappeared and the interior also became solid after crystallization, while the morphology of the hexahedron was basically maintained. In high-resolution TEM (HRTEM) image of C-FeNi, (111) and (222) crystal faces of $NiFe_2O_4$ and (200) crystal face of Ni can be observed (Figure S6, Supporting Information). High-angle annular dark-field scanning transmission electron microscopy (HAADF-STEM) observation of A-FeNi was performed as shown in Figure 2F, Figures S7, and S8 (Supporting Information). The STEM mapping and line scanning results illustrated that elements Fe, Ni, C, N, O were all distributed uniformly in amorphous A-FeNi electrocatalyst.

The crystallinity of the as prepared Ni-based MOFs was analyzed by powder X-ray diffraction (PXRD) and the experimental PXRD patterns of Ni-MOF, FeNi-MOF, CoNi-MOF, and

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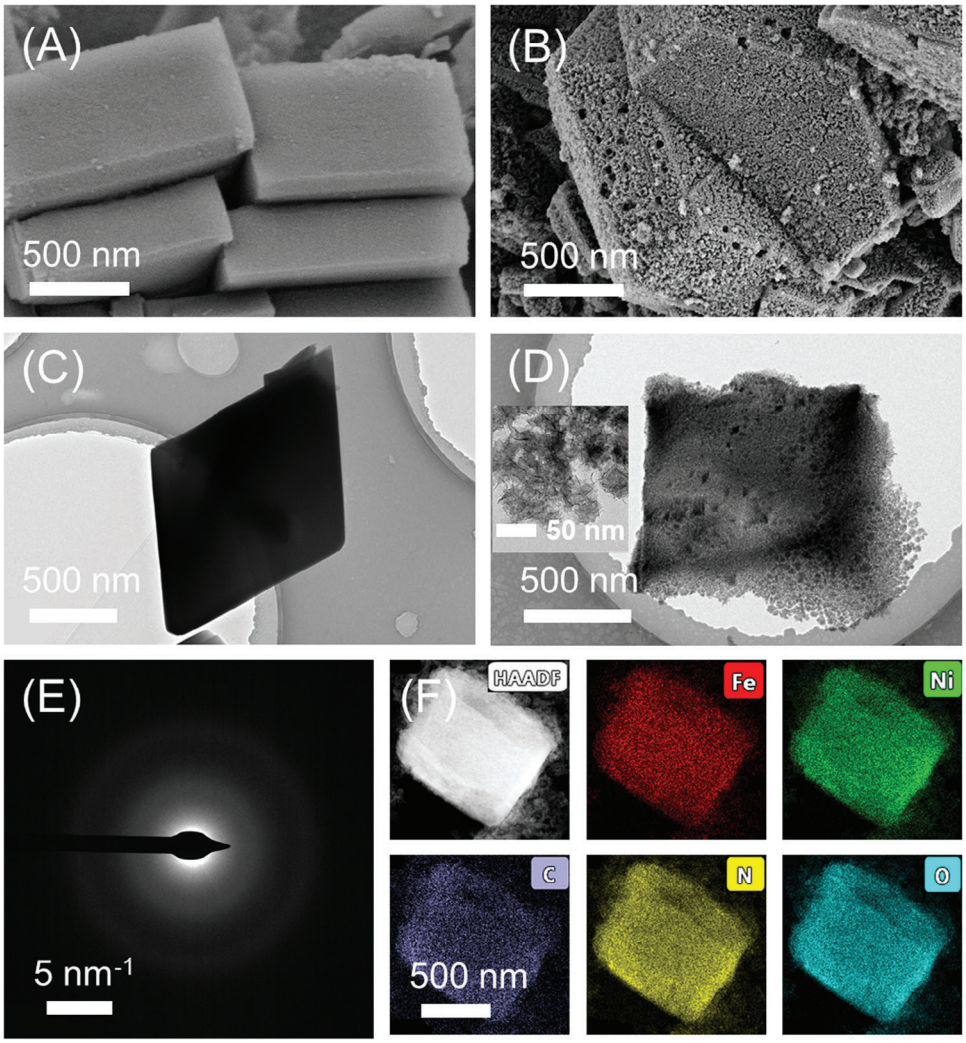


Figure 2. Morphology studies. A,B) SEM images of FeNi-MOF, A-FeNi. C,D) TEM images of FeNi-MOF, A-FeNi. E) SAED pattern of A-FeNi. F) HAADF-STEM image and the corresponding STEM EDS mappings of A-FeNi.

MnNi-MOF corresponded very well with the simulated pattern as shown in **Figures** 3A and S9. The structural diagram of FeNi-MOF is shown in Figures 1 and S10 (Supporting Information), metal ions were coordinated with 4 nitrogen atoms and 2 oxygen atoms. After the in situ conversion of MOFs into catalysts in the electrolyte, the catalysts exhibited an amorphous form as show in Figures 2E and 3B. The obtained amorphous catalyst showed porous morphology on both surface and internal, and its Brunauer-Emmett-Teller (BET) specific surface area was characterized to be 117 $m^2\ g^{-1}$ as shown in Figure 3C. Moreover, Barret-Joyner-Halenda (BJH) pore size distribution was conducted for A-FeNi (inset in Figure 3C). The average pore size was 8.8 nm and the pore volume was calculated to be 0.27 mL g^{-1}. On the surface of this series of MOFs, both Ni^{2+} and Ni^{3+} can be detected simultaneously in Ni 2p X-ray adsorption spectroscopy (XPS) spectra as shown in Figure S11 (Supporting Information). When part of the Ni ions were replaced by other metals (Fe, Mn, Co), the binding energy of Ni 2p in bimetallic MOF decreased. Ni in MOF-derived amorphous catalysts was mainly in the form of Ni^{2+} as shown in Figure 3D and Figure S12 (Supporting Information).[9] Compared with the monometallic Ni-MOF, the binding energy of Ni 2p in the amorphous bimetallic catalysts was negatively shifted, with the largest negative shift occurring in Ni 2p of A-FeNi as shown in Figure S12 (Supporting Information). Compared to FeNi-MOF, Fe 2p binding energy shifted positively in A-FeNi as shown in Figure S13 (Supporting Information), indicating that Fe^{2+} was oxidized to Fe^{3+} during the amorphous transition. Same situation existed in the transformation of CoNi-MOF to A-CoNi, as shown in Co 2p XPS spectra (Figure S14, Supporting Information). The amorphous catalyst was annealed to obtain the crystallization catalyst C-FeNi. The PXRD peaks of C-FeNi can be corresponded to two species including Ni and $NiFe_2O_4$, while Ni and NiO can be detected in C-Ni (Figure S15, Supporting Information). After annealing, Ni 2p XPS indicated the presence of Ni on the surface of C-FeNi as shown in Figure 3D, which corresponded to the PXRD result. The deconvolution of Fe 2p for A-FeNi and C-FeNi was illustrated in Figure 3E. The binding energy of Fe 2p in C-FeNi negatively shifted compared to which

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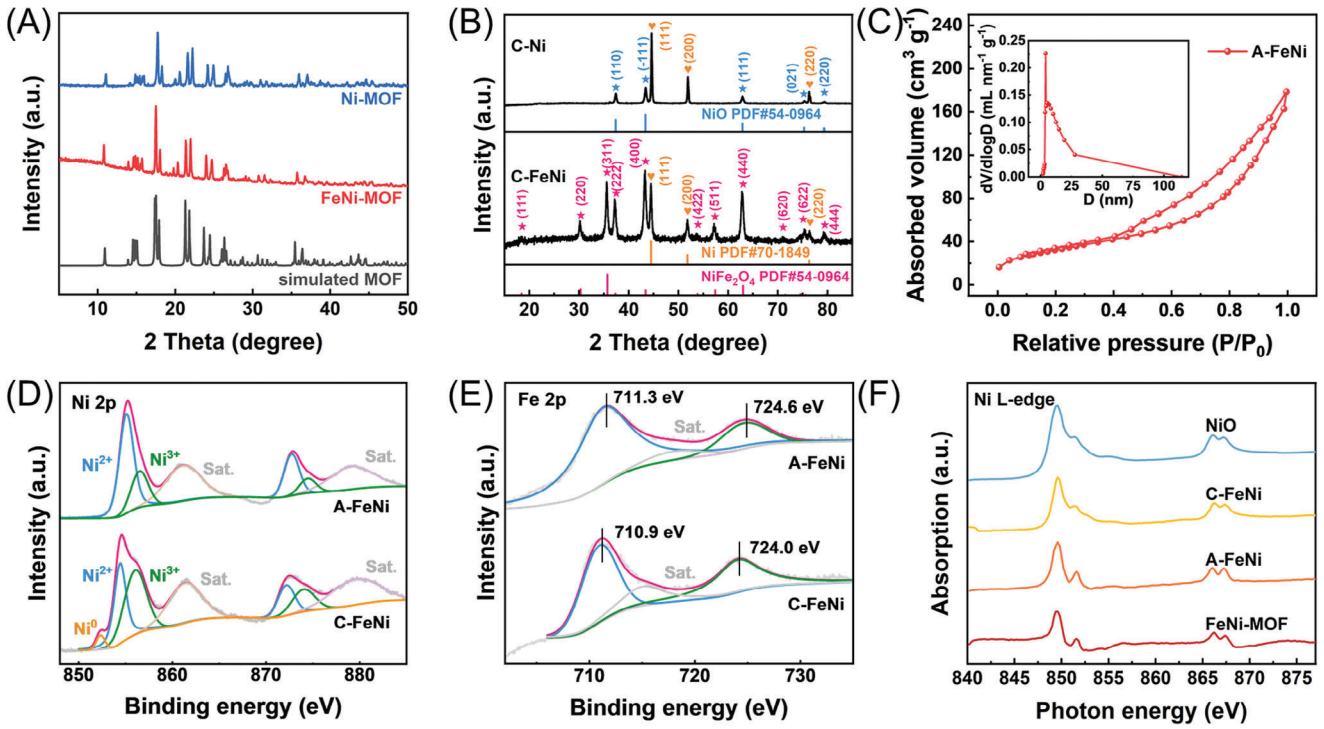


Figure 3. Structure characterization. A) The simulated XRD pattern and PXRD patterns of Ni-MOF and FeNi-MOF. B) PXRD patterns of A-FeNi. C) BET Isothermal curve of N_2 adsorption and desorption and BJH pore size distribution of A-FeNi. D) Ni 2p spectra of A-FeNi and C-FeNi. E) Fe 2p spectra of A-FeNi and C-FeNi. F) Ni L-edge XANES of FeNi-MOF, A-FeNi, C-FeNi, and NiO.

of A-FeNi due to the reduction of Fe^{3+} in Ar atmosphere during annealing. The deconvolution of A-FeNi O 1s spectra revealed the coexistence of metal–oxygen bond (O1, 530.2 eV), oxygen vacancy (O2, 531.3 eV), and carbon–oxygen bond (O3, 532.6 eV) as shown in Figure S16A,B (Supporting Information).[33–35] Oxygen vacancy (O2 peak) accounted for the maximum proportion higher than which of FeNi-MOF, demonstrating a large number of oxygen vacancies were generated during the amorphization process. The C 1s peaks can be assigned to C–C (284.8 eV), C–N (288.4 eV), and N–$(C)_3$ (288.7 eV),[36] corresponding to hexamethylenetetramine as shown in Figure S17 (Supporting Information). N 1s spectra can be divided into two peaks including C–N (398.4 eV) and metal–N (399.3 eV), and the peak of metal–N was originated from the coordination bond between metal ions and N atoms (Figure S18, Supporting Information).[37–39] In the X-ray absorption near edge structure (XANES) spectra conducted in Shanghai Synchrotron Radiation Facility (SSRF) beamline BL08U, it can be observed that the spectrum of amorphous catalyst A-FeNi was similar to which of FeNi-MOF, indicating that they had similar structures, while the spectrum of crystallized catalyst C-FeNi is closer to standard NiO due to the presence of metal oxide $NiFe_2O_4$ is one of the main components of C-FeNi. Fourier transform-infrared reflection (FT-IR) characterization of MOFs and MOFs-derived catalysts was conducted as shown in Figure S19 (Supporting Information). Peaks for ligand were detected on the IR spectrum of precursor MOFs, and small amount of ligand peaks were still detected even after the derivation into amorphous catalysts. The Raman results were similar to the FT-IR results as shown in Figure S20 (Supporting Information), the ligand peaks could be detected in both amorphous catalysts and precursor MOFs, indicating the presence of undestroyed ligand in the amorphous catalysts.[40–42] The FT-IR and Raman spectra of this series of precursor MOFs and catalysts are basically the same, indicating that the series of amorphous catalysts have similar coordination environment as shown in Figures S21 and S22 (Supporting Information).

K-edge X-ray absorption fine structure (XAFS) spectroscopy was carried out in order to deeply investigate the coordination environment and electronic structure of transition metals. Ni K-edge XAFS spectra were conducted as shown in **Figure** 4A, the average valance state of metal can be roughly obtained by linear fitting after differentiation of XANES as shown in Figure S23 (Supporting Information). The valence for Fe in FeNi-MOF and A-FeNi was 2.5 and 3.4, and the valance for Ni in FeNi-MOF and A-FeNi was 2.4 and 2.6. Figure 4B exhibited Fourier transformed extended X-ray absorption fine structure (EXAFS) in *R*-space, and there were two primary peaks for Ni K-edge in FeNi-MOF, A-FeNi, and NiO at 1.5 and 2.7 Å. In regard to Fe K-edge XAFS for FeNi-MOF, A-FeNi, and Fe_2O_3 in R-space, two peaks located at 1.4 and 2.8 Å can be observed, which both represented Fe–O or Fe–N. Structural fitting of EXAFS can yield more information about local atom structure. The fitted curve is in good agreement with the experimental curve in both R-space and q-space for FeNi-MOF and A-FeNi as shown in Figures S24 and S25 (Supporting Information). The average bond length (R), coordination number (CN, amp), and disorder were obtained by EXAFS fitting. The fitting results are demonstrated in Table S2 (Supporting Information), the CN of both Fe and Ni in FeNi-MOF was ≈4–5, which

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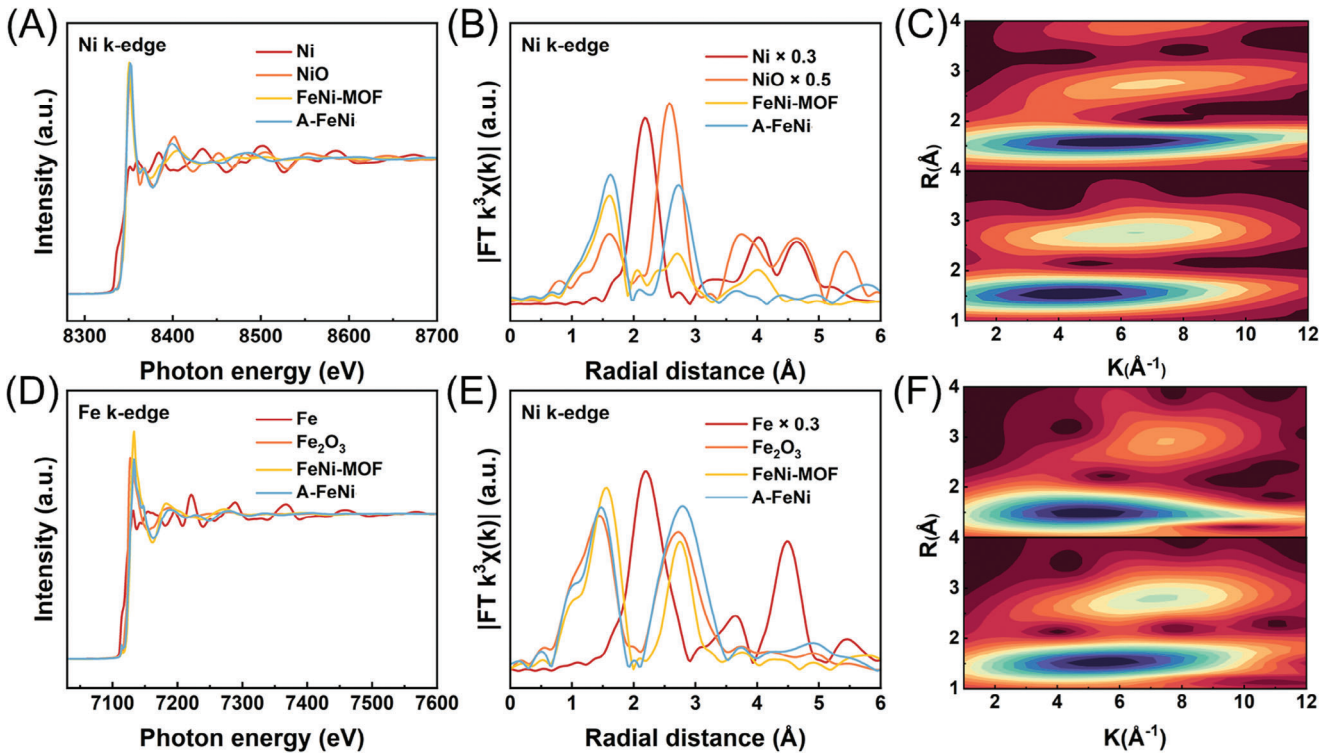


Figure 4. EXAFS studies. A) Ni K-edge XAFS spectra of Ni foil, NiO, FeNi-MOF, and A-FeNi. B) Fourier transformed Ni K-edge EXAFS spectra of Ni foil, NiO, FeNi-MOF, and A-FeNi. C) Ni K-edge WT-XAFS of FeNi-MOF and A-FeNi. D) Fe K-edge XAFS spectra of Fe foil, Fe_2O_3, FeNi-MOF, and A-FeNi. E) Fourier transformed Ni K-edge EXAFS spectra of Fe foil, Fe_2O_3, FeNi-MOF, and A-FeNi. F) Fe K-edge WT-XAFS of FeNi-MOF and A-FeNi.

resulted from the loss of ligand H_2O that was easily shed in precursor MOFs. In addition, the coordination bond lengths for Ni–N(O) and Fe–N(O) were 2.12 ± 0.01 Å and 2.10 ± 0.02 Å, respectively. As for A-FeNi, the CN for both Fe and Ni in A-FeNi was ≈6, and the coordination bond lengths for Ni–N(O) and Fe–N(O) were fitted to be 2.08 ± 0.01 Å and 2.08 ± 0.01 Å, respectively. The above results indicated that after the in situ transformation of FeNi-MOF into amorphous A-FeNi, the coordination structure in FeNi-MOF is partially preserved, while the metal ions developed a higher coordination. Moreover, wavelet transform XAFS (WT-XAFS) results were shown in Figure 4C,F for Ni K-edge and Fe K-edge, it can be much more intuitively observed from the bright spots in WT-XAFS that FeNi-MOF and A-FeNi had similar local electronic structures. The six-coordination state and the inherited ligands may be one of the reasons for the ultra-high performance of the amorphous A-FeNi.[43,44] The inherited ligands in A-FeNi included C≡N and ─CNC according to FT-IR and Raman results. Ligands in A-FeNi inherited from FeNi-MOF can maintain the coordination environment of the metal center to optimize the OER activities. The interaction between organic ligands and matel sites can affect diffusion rates within the pores, therefore influence the transport rates of reactants and products to and from the active sites modulating OER rates.

OER performance of the catalysts was first characterized in 1 M KOH solution using a three-electrode system at 25 °C, and reversible hydrogen electrode was applied as reference electrode. **Figure** 5A exhibits the linear sweep voltammetry (LSV) curves of A-FeNi, C-FeNi, A-Ni, C-Ni, RuO_2, and pure Ni foam, respectively. Amorphous monometallic A-Ni showed a relatively poor OER performance, with an overpotential of 188 mV at 10 mA cm^{-2}. While a portion of Ni was replaced by Fe, A-FeNi exhibited the most excellent OER catalytic performance of this series, with an overpotential of 152 mV at 10 mA cm^{-2} and 232 mV at 100 mA cm^{-2}. In addition, powder sample on carbon paper was also tested for OER performance as shown in Figure S26 (Supporting Information) to prevent the interaction between the nickel foam and the catalyst from affecting the performance. The performance trends obtained were the same as those in Figure 5A. Moreover, the overpotentials for A-CoNi, A-MnNi were 198 and 275 mV as shown in Figure S27 (Supporting Information). When A-FeNi is annealed and transformed to C-FeNi, C-FeNi needed 219 mV to achieve a current density of 10 mA cm^{-2}, which is much poorer than that of A-FeNi. Besides, C-Ni needed an overpotential of 252 mV at 10 mA cm^{-2}. The crystalline monometallic catalyst has the worst performance in this series. Moreover, in the case of similar loading, the OER performance of A-FeNi is far better than that of commercial RuO_2. The performance of the catalyst was also tested in simulated seawater as shown in Figure S28 (Supporting Information). The results demonstrated that the catalysts maintain OER activity and exhibit a satisfactory performance in seawater environments. The Tafel slope was conducted as shown in Figure 5B to represent the kinetic velocity from 1 to 10 mA cm^{-2}. A-FeNi showed the smallest Tafel slope of 17 mV dec^{-1} among those of the series, which demonstrated the fastest dynamic speed. The Tafel slope for A-Ni, C-FeNi, C-Ni, commercial RuO_2, and Ni foam is 33, 42, 51, 45, and 102 mV dec^{-1}.

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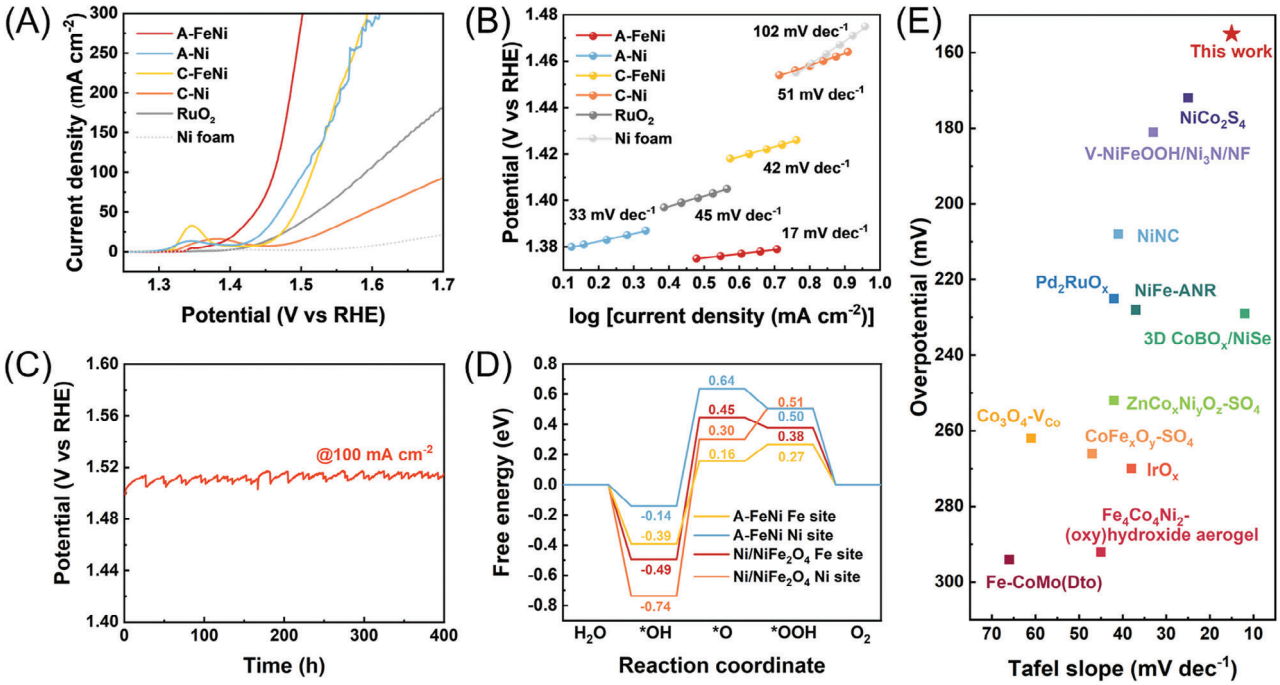


Figure 5. Electrochemical measurement in 1 M KOH and DFT test. A) A-FeNi, C-FeNi, A-Ni, C-Ni, RuO_2, and Ni foam LSV curves towards OER. B) Tafel slope for A-FeNi, C-FeNi, A-Ni, C-Ni, RuO_2, and Ni foam. C) Durability test for A-FeNi at 100 mA cm^{-2}. D) Gibbs free energy diagram for A-FeNi and C-FeNi at Ni and Fe site respectively. E) Comparison of the OER electrocatalytic properties of A-FeNi to other amorphous catalysts.

The kinetic velocity of A-FeNi is much better than the other catalysts in this series and also commercial RuO_2, which endowed A-FeNi with superior OER activity. A-FeNi has the most advanced OER performance compared to other MOFs-related amorphous (or amorphous-containing) catalysts as shown in Figure 5E and Table S3 (Supporting Information). Electrochemical impedance spectroscopy (EIS) was used to estimate the charge transfer resistance (R_{ct}) of the catalyst, as shown in Figure S29 (Supporting Information). A-FeNi had minimal impedance among the catalysts, which assigned the highest electron conduction efficiency to A-FeNi. In addition, A-FeNi has excellent durability and can work continuously for at least 400 h at high current density 100 mA cm^{-2} (Figure 5C), which is of great value for the application under practical circumstance. In order to further explore the stability of the materials, the morphology and crystalline state of the A-FeNi after OER were characterized as shown in Figure S30 (Supporting Information), and the results showed that A-FeNi was able to maintain the porous cubic morphology and remain amorphous. DFT calculation was performed to gain further insight into the OER electrocatalytic mechanism of A-FeNi following $4e^-$ mechanism proposed by Nørskov as shown in Figure 5D.[45] During the first step of the OER process, Ni active sites in A-FeNi performed a critical role on reducing the adsorption energy. While in the next three steps in OER process, Fe active sites in A-FeNi acted as primary active sites according to the change of Gibbs free energy.

3. Conclusion

In summary, we developed a highly efficient MOFs-derived amorphous electrocatalyst A-FeNi for OER. Bimetallic A-FeNi exhibited porous hexahedron morphology and can be solidly and rapidly loaded on Ni foam under an extremely mild condition. The obtained A-FeNi exhibited a similar morphology and structure to the precursor FeNi-MOF. The properties of A-FeNi were far superior to that of crystalline-stated C-FeNi since the amorphous structure of A-FeNi is capable of maintaining partial ligand and coordination, and also preventing phase separation. A-FeNi only needed an overpotential of 152 mV at 10 mA cm^{-2} with a Tafel slope of 17 mV dec^{-1} for OER, which is one of the best compared to recently reported literature. Moreover, A-FeNi can achieve scalable preparation and excellent durability. The characterization results showed that amorphous and bimetallic structure facilitated the modulation of the electronic structure and Gibbs free energy to achieve optimal OER performance. In this work, a series of Ni-based MOFs-derived amorphous catalysts were innovatively obtained using a facile and scalable method with first-rate OER activity, and this work is expected to bring new ideas to the design of MOFs-related electrocatalysts.

4. Experimental Section

Chemicals: $Ni(BF_4)_2{\cdot}6H_2O$, Nafion solution (5 wt.%) and RuO_2 were purchased from Alfa Aesar. $NaN(CN)_2$, KOH, $Fe(ClO_4)_2{\cdot}xH_2O$, $Co(NO_3)_2{\cdot}6H_2O$ and $MnCl_2$ were purchased from Energy Chemical. Hexamethylenetetramine was purchased from Sinopharm Chemical. All the reagents were used without any further purification. Millipore water (≈18 MΩ cm) was used throughout the experiment.

Synthesis of FeNi-MOF: Ni foam was first cut into 2×3 cm^2 pieces and washed with acetone, 1 M HCl, deionized water, and ethanol. 0.75 mmol $Fe(ClO_4)_2{\cdot}xH_2O$ and 0.75 mmol $Ni(BF_4)_2{\cdot}6H_2O$ were dissolved in 10 mL methanol in 50 mL beaker, then immersed one piece of treated dry Ni

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foam in the above solution to form A. 1.50 mmol hexamethylenetetramine was dissolved in 5 mL methanol to form B. 3.00 mmol $NaN(CN)_2$ was dissolved in 5 mL H_2O to form C. Then mixed B and C and dropped the mixture into A. After leaving it for 1 h, the green FeNi-MOF was firmly grown on Ni foam. The composite electrode was ultrasonically washed with methanol for four times and then dried under vacuum at 50 °C.

Synthesis of Other MOFs: Ni-MOF was synthesized by 1.5 mmol $Ni(BF_4)_2{\cdot}6H_2O$ with other conditions unchanged. CoNi-MOF was synthesized by 0.75 mmol $Co(NO_3)_2{\cdot}6H_2O$ and 0.75 mmol $Ni(BF_4)_2{\cdot}6H_2O$ with other condition unchanged. MnNi-MOF was synthesized by 0.75 mmol $MnCl_2$ and 0.75 mmol $Ni(BF_4)_2{\cdot}6H_2O$ with other condition unchanged.

Synthesis of A-FeNi: Before OER, FeNi-MOF was immersed in 1 M KOH electrolyte, and the MOF was immediately converted into A-FeNi.

Synthesis of Other Amorphous Electrocatalysts: A-Ni, A-CoNi, and A-MnNi were synthesized by the same method as A-FeNi.

Synthesis of C-FeNi: The obtained A-FeNi was calcined in a tube furnace under Ar flow at 500 °C for 2 h (heating rate: 10 °C min^{-1}) to obtain C-FeNi.

Synthesis of Other Crystallized Electrocatalysts: C-Ni, C-CoNi, and C-MnNi were synthesized by the same method as C-FeNi.

Large Scale Synthesis of FeNi-MOF: 343 g $Fe(ClO_4)_2{\cdot}xH_2O$ and 426 g $Ni(BF_4)_2{\cdot}6H_2O$ were dissolved in 12.5 L methanol. Three hundred fifty grams of hexamethylenetetramine was dissolved in 6.3 L methanol. Mix the above two solutions uniformly. Four hundred forty five grams of $NaN(CN)_2$ was dissolved in 6.3 L H_2O and the obtained solution was slowly added to the mixture above under stirring for 2 h. The large-scale FeNi-MOF sample was collected by suction filtration and washed with water.

Characterization: Powder X-ray diffraction patterns were conducted on Shimadzu, XRD-7000 with Cu Kα radiation. The morphology of the catalysts was collected through SEM (Carl Zeiss, Ultra-55, and JEOL, JSM-IT800), TEM (JEOL, JEM-2100, and JEM-F200), and HAADF-STEM (FEI, TALOS F200). XPS was measured on Thermo Scientific K-Alpha with Al Kα radiation and rectified by C 1s = 284.80 eV. The FT-IR spectra were characterized on Thermo Scientific Nicolet iS20. The Raman spectra were conducted on Horiba LabRAM HR Evolution. The ICP-MS test was conducted on Thermo Scientific iCAP 6300. BET Surface area and porosity were tested at Beishide, BSD-PS1. The Ni, Fe, Co, Mn L-edge XANES data were collected on BL08U at Shanghai Synchrotron Radiation Laboratory (SSRL) and the Ni, Fe, Co, Mn K-edge XAFS data was collected on BL11B at SSRL.

Electrochemical Measurement: The electrocatalysis performance was characterized by a Princeton PARSTAT 3000A-DX workstation with a classic three-electrode system in 1 M KOH electrolyte. A reversible hydrogen electrode was applied as reference electrode and a graphite rod (98%) was applied as counter electrode. The catalyst loading on Ni foam was cut into 1×1 cm^2 and applied as working electrode (loading mass ~5 mg cm^{-2}). RuO_2 was also loaded on 1×1 cm^2 Ni foam using Nafion solution with loading mass of also ≈5 mg cm^{-2}. The LSV curves were obtained at a scan rate of 2 mV s^{-1} with 95% iR compensation. The electrochemical impendence spectroscopy was performed from 10^5 to 10^{-2} Hz with the potential of 0.1 V versus open circuit (OC).

Computational Method: Structure relaxation and electronic energy calculations were performed with spin-polarized density functional theory as implemented in the Vienna Ab-initio Simulation Package (VASP) code. In the process of calculations, the projector augmented-wave (PAW) technique was applied and the plane-wave energy cut-off was set to 350 eV. Perdew–Burke–Ernzerhof functional was used to treat the correlation-exchange energies of the systems. The sampling over Brillouin zone was treated by a $(6 \times 2 \times 1)$ Monkhorst-Pack grid. The vacuum layer was set to 15 Å to avoid the interaction of periodic structures. In addition, considering the calculation of adsorption energy, vdW3 correlation (vdW-DF3) was used to explain the van der Waals interactions. Geometry optimization was repeated until the total energy tolerance was converged to 5×10^{-4} eV and the changes of the force on the atoms $<5 \times 10^{-3}$ eV $Å^{-1}$. To estimate the OER property of the electrocatalysts, the Gibbs free energy of coordinate elementary steps were calculated based on the following $4e^-$ mechanism (1a-1d) proposed by Nørskov.

$$H_2O\ (l) + * \leftrightharpoons * OH + H^+ + e^- \quad (1a)$$

$$* OH \leftrightharpoons * O + H^+ + e^- \quad (1b)$$

$$* O + H_2O\ (l) \leftrightharpoons * OOH + H^+ + e^- \quad (1c)$$

$$* OOH \leftrightharpoons * + O_2\ (g) + H^+ + e^- \quad (1d)$$

where "*" denotes the active sites on the catalyst, (*l*) and (*g*) represents liquid and gas phases, respectively.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

amorphous, electrocatalysis, metal-organic frameworks, oxygen evolution reaction, scalable synthesis

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